

In the figure shown above, what is the maximum number of different ways in which 8 identical balls can be placed in the small triangles 1, 2, 3 and 4 such that each triangle contains at least one ball?

(a) 32 (b) 35
(c) 44 (d) 56

Ans. (b) : There can be 5 cases–

1. First triangle can have 5 balls and rest three 1 each

   So no. of ways for $(5,1,1,1) = \frac{4!}{3!} = 4$

2. First triangle can have 4 balls, 2nd can have 2 balls and rest 1 each

   So no. of ways for $(4,2,1,1) = \frac{4!}{2!} = 12$

3. Similarly no. of ways for combination

   $(2,2,2,2) = \frac{4!}{4!} = 1$

4. No. of ways for $(3,3,1,1) = \frac{4!}{2!\times 2!} = 6$

5. No. of ways for $(3,2,2,1) = \frac{4!}{2!} = 12$

As either of these 5 cases are possible, Hence total no. of ways = 4 + 12 + 1 + 6 + 12 = 35

146 Three dice (each having six faces with each face having one number from 1 to 6) are rolled. What is the number of possible outcomes such that at least one dice shows the number 2?

(a) 36 (b) 81
(c) 91 (d) 116

Ans. (c) : Total number of outcome when 3 dice are rolled = 6 × 6 × 6 = 216
Number of way in which 2 does not appear on any of the dice = 5 × 5 × 5
= 125
So, the required possibility to show 2 atleast one time will be = 216 – 125 = 91

147. All the six letters of the name SACHIN are arranged to form different words without repeating any letter in any one word. The words so formed are then arranged as in a dictionary. What will be the position on the word SACHIN in the sequence?

(a) 436 (b) 590
(c) 601 (d) 751

Ans. (c) : In S A C H I N
S is fixed since it is the starting letter
No. of words that start with A
=A–––– =5×4×3×2×1 =5! =120
No. of words starting with C
=C––––– =5!=120
Similarly, no. of words starting from H,I,N will be 120
∴ Total No. of words=120+120+120+120+120=600
Since SACHIN will be ranked after 600 words,
therefore it's rank will be 601th .

148. Five balls of different colours are to be placed in three different boxes such that any box contains at least one ball. What is the maximum number of different ways in which this can be done?

(a) 90 (b) 120
(c) 150 (d) 180

Ans. (c) : Let the boxes be x, y, z
I – x, y, z : 1, 1, 3
Total 3 case possible – $3\times {}^5C_3\ {}^2C_1\ {}^1C_1 = 3\times 10\times 2 = 60$
II – x, y, z : 1, 2, 2
Total 3 case possible – $3\times {}^5C_2\ {}^3C_2\ {}^1C_1 = 90$
Total ways = I + II = 60 + 90 = 150

149. Amit has five friends : 3 girls and 2 boys. Amit's wife also has 5 friends: 3 boys and 2 girls. In how many maximum number of different ways can they invite 2 boys and 2 girls such that two of them are Amit's friends and two are this wife's?

(a) 24 (b) 38
(c) 46 (d) 58

Ans. (c) :
2 boys from Amit friend and 2 girls from his wife friend
OR 1 boy & 1 girl from Amit friend and 1 boy and 1 girl from his wife friend
OR 2 girls from Amit friend and 2 boys from his wife friend
Number of ways =1×1+3×2×3×2+3×3=1+36+9 = 46

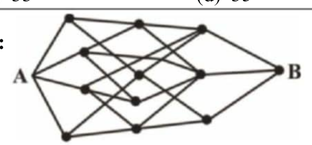
150.



(Each small circle represents a different station) What is the maximum number of different paths that exist between the station A and the station B?

(a) 28 (b) 31
(c) 33 (d) 35

Ans. (b) :



There are 4 roots between A to b vio P, Q, R and S
Case 1 : rout via P →
A to P = 3 and P to B = 3
Routes via P = 3 × 3 = 9
Case 2 : A to Q = 4 and Q to B = 3
Routes via Q = 4 × 3 = 12
Case 3 : Similarly routes via R = 3 × 2 = 6
Case 4 : Routes via S = 2 × 2 = 4
Total no. of routes = 9 + 12 + 6 + 4 = 31

UNION PUBLIC SERVICE COMMISSION

Civil Services (Preliminary Exam) - 2006

GENERAL STUDIES : PAPER-I

Time: 2 hours **Maximum Number: 200**

ANCIENT HISTORY

1. The Allahabad Pillar inscription is associated with which one of the following?

(a) Mahapadma Nanda
(b) Chandragupta Maurya
(c) Ashoka
(d) Samudragupta

Ans. (d) : The Allahabad Pillar inscription is associated with Samudragupta's court poet Harisena.

2. Between which of the following was the ancient town of Takshsila located?

(a) Indus and Jhelum (b) Jhelum and Chenab
(c) Chenab and Ravi (d) Ravi and Beas

Ans. (a) : Taxila was located between River Indus and Hydaspes, river Hydaspes is now known as river Jhelum, it is a tributary of river Indus. It is located in Rawalpindi,Punjab.

3. With reference to the invaders in ancient India, which one of the following is the correct chronological order?

(a) Greeks-Sakas-Kushans
(b) Greeks-Kushans-Sakas
(c) Sakas-Greeks-Kushans
(d) Sakas-Kushans-Greeks

Ans. (a) : The invasion in ancient India was firstly done by Greeks, then Sakas and then Kushanas. The Greek conquests of India took place in the years before the Common Era, and a rich trade flourished between India and Greece, especially in silk, spices, and gold. The Greeks invaded India several times, starting with the conquest of Alexander the Great between the years 327 to 326 B.C.E. The Indo-Greek rule lasted from about 180 BC till about 55 BC. The Sakas were alternatively known as Indo-Scythians, invaded northwest India in the first century BC onwards. The Kushan Empire fragmented into semi-independent kingdoms in the 3rd century AD, which fell to the Sasanians invading from the west, establishing the Kushano-Sasanian Kingdom in the areas of Sogdiana, Bactria and Gandhara. In the 4th century, the Guptas, an Indian dynasty also pressed from the east.

4. Consider the following statements:

1. The Ikshvaku rulers of Southern India were antagonistic towards Buddhism.
2. The Pala rulers of Eastern India were patrons of Buddhism.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b) : Most of the inscriptions of the Andhra Ikshvaku period record either the construction of the Buddhist viharas or the gifts made to them which show that the Ikshvaku rulers of Southern India were supporters of Buddhism. The Pala rulers of Eastern India were patrons of Buddhism. They established famous Vikramshila vihar.

5. Who among the following laid the foundation of Rashtrakuta Empire?

(a) Amoghavarsha I (b) Dantidurga
(c) Dhruva (d) Krishna I

Ans. (b) : Dantivarman or Dantidurga was the founder of the Rashtrakuta dynasty. The dynasty ruled large parts of the Indian subcontinent between the sixth and 10th centuries. Dantidurga was a feudatory of the Chalukyas. In the mid-8th century, Dantidurga rose to importance and established himself in place of the declining Chalukya dynasty.

6. Which one of the following pairs is correctly matched?

(a) Vikramsila Monastery : Uttar Pradesh
(b) Hemkund Gurudwara : Himachal
(c) Udayagiri : Maharashtra
(d) Amaravati Buddhist Stupa : Andhra Pradesh

Ans. (d) : Amaravati Buddhist Stupa : Andhra Pradesh

7. Match List-I (Finding/Invention/Calculation) with List-II (Ancient Indian Scholar) and select the correct answer using the code given below the Lists:

List-I (Finding/Invention /Calculation)	List-II (Ancient Indian Scholar)
A. Time taken by the Earth to orbit the Sun	1. Aryabhatta
B. Calculation of the value of π ('pi')	2. Bhaskaracharya
C. Invention of the digit zero	3. Budhayana
D. The game of snakes and ladders	4. Gyandev

Codes:

	A	B	C	D
(a)	2	4	1	3
(b)	1	3	2	4
(c)	2	3	1	4
(d)	1	4	2	3

Ans. (c) : A-2 , B-3, C- 1, D- 4